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ITAI 3377 – Edge & IIoT

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A04 Individual Reflective Journal – Lab 04

February 16, 2026

Reflective Journal: IIoT Protocols Project

Introduction

In Lab 04, I worked on understanding Industrial Internet of Things communication protocols, specifically MQTT, CoAP, and OPC UA. This lab connected directly to what we learned in Module 04 about IIoT networking and how devices communicate in industrial systems. The main purpose of the project was to compare these protocols and understand when and why each one would be used.

Going into this lab, my goal was to really understand how communication works behind the scenes in smart factories and industrial automation. I've heard these protocol names before, but I wanted to understand what actually makes them different and how they impact performance, security, and reliability.

Personal Contributions

For this project, I focused on breaking down how each protocol works and explaining their real-world use cases. I worked on analyzing MQTT's publish/subscribe model and how it relies on a broker to distribute messages. I also reviewed how CoAP works using a REST-style approach over UDP and how that affects speed and reliability.

I spent time comparing the strengths and weaknesses of all three protocols and helped organize the comparison section so it clearly showed when each protocol would be most appropriate. I made sure the explanations were simple but still technically accurate.

Learning Outcomes

This lab helped me understand that not all communication protocols are built the same, even if they are all used in IoT environments.

With MQTT, I learned that it is lightweight and designed for devices with limited power and bandwidth. The publish/subscribe system makes it efficient because devices don't have to communicate directly with each other. The different Quality of Service levels also help balance reliability and performance.

With CoAP, I learned that it is UDP-based and designed for constrained environments. It is simpler than HTTP and works well for devices that need fast communication but can tolerate some packet loss. I now understand why CoAP is good for resource discovery and observation in smart systems.

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With OPC UA, I realized it is more advanced and built specifically for industrial automation. It supports strong security, structured data modeling, and interoperability between vendors. Compared to MQTT and CoAP, it is more complex but also more powerful for large industrial systems.

One big takeaway for me was that protocol selection depends on system requirements like latency, scalability, and security. There is no “one-size-fits-all” solution.

Challenges and Solutions

One challenge I had was fully understanding the difference between MQTT and CoAP at first. Both are lightweight and used in IoT systems, so they seemed very similar. After reviewing how MQTT depends on a broker and how CoAP uses direct request/response communication, the differences became clearer.

Another challenge was understanding OPC UA’s data modeling structure because it is more detailed and industrial-focused. I had to go back to the lecture slides and review examples until it made sense how it organizes and secures data.

Working through those challenges helped me understand the material more deeply instead of just memorizing definitions.

Future Applications

The knowledge I gained from this lab will definitely help in future projects involving IIoT, edge computing, or smart industrial systems. Understanding how these protocols work will allow me to design systems that are efficient and secure.

If I were to improve this project, I would like to test these protocols in a simulated environment to measure latency and reliability differences. That hands-on testing would make the learning even stronger.

Overall, this lab helped me better understand how communication protocols form the backbone of IIoT systems and why choosing the right one matters.